Advances in Soft X-Ray Beamline Design: Grating Monochromator Simulations

Theory, Aberration Analysis, and Ray-Tracing Examples

Based on lectures by **Ruben Reininger**
(APS/ANL), originally presented at the
Software for Optical Simulations Workshop,
Trieste, October 2016;
adapted by M. Sanchez del Rio,
DESY Hamburg, May 2025

April 15, 2026

Contents

1 Introduction and Scope **2**

2 Theoretical Foundation: Optical Path Length and Fermat's Principle **2**
2.1 The Optical Path Length Function . . . 2
2.2 Meaning of the Coefficients . . . 3
2.3 Grating Equation and Resolving Power . . . 3
2.4 Image Size: Sanity Check Formula . . . 4

3 Spherical Grating Monochromator (SGM) **4**
3.1 Optical Layout . . . 4
3.2 Ray-Tracing Example at 1000 eV . . . 5
3.3 Ray-Tracing Example at 1600 eV . . . 5

4 Plane Grating Monochromator (PGM) **6**
4.1 Key Properties of a Plane Grating . . . 6
4.2 Collimated PGM: Vertical Deflection Layout . . . 7
4.3 Follath Collimated PGM: Horizontal Deflection . . . 8

5 Focusing Variable Line Spacing PGM (FVLS-PGM) **9**
5.1 Principle . . . 9
5.2 Optical Layout . . . 10
5.3 Heat Load and Correction . . . 10

6 Comparison of Monochromator Types **12**

7 Summary and Conclusions **12**

Glossary of Symbols **14**

References **14**

How to Replace Placeholders **15**

1. Introduction and Scope

This document summarises the theoretical framework and ray-tracing simulation examples presented by Reininger for soft X-ray (SXR) grating monochromators. Four main instrument types are treated in order of increasing optical sophistication:

1. Spherical Grating Monochromator (SGM)
2. Vertically collimated Plane Grating Monochromator (vertical PGM)
3. Follath collimated PGM (horizontal deflection, fixed focal plane)
4. Focusing Variable Line Spacing PGM (FVLS-PGM)

All simulations use the OASYS/SHADOW ray-tracing environment. Unless otherwise stated, the source parameters correspond to the (then) present APS undulator: a 2 m in-vacuum device with realistic emittance, divergence, and energy spread.

Fig. 1

Overview schematic of the four monochromator types covered in this document (SGM, vertical PGM, Follath PGM, FVLS-PGM), showing their relative complexity and key optical components.

2. Theoretical Foundation: Optical Path Length and Fermat's Principle

2.1. The Optical Path Length Function

The foundation for all grating monochromator designs is the **optical path length function** F for a toroidal grating, expanded in terms of the sagittal coordinate w (across the groove direction) and the meridional coordinate ℓ (along the grooves):

$$F = F_{00} + F_{01}\, w + \tfrac{1}{2} F_{20}\, w^2 + \tfrac{1}{2} F_{02}\, \ell^2 + \tfrac{1}{2} F_{30}\, w^3 + \tfrac{1}{2} F_{21}\, w\, \ell^2 + \cdots \tag{1}$$

Fermat's principle requires that, for each diffracted order, $\delta F = 0$ with respect to variations in the ray coordinates. Setting each aberration coefficient to zero yields the design conditions for a perfect (aberration-free) monochromator.

Fig. 2

Geometry of a toroidal grating showing source point S, grating vertex, image point S', sagittal coordinate w, meridional coordinate ℓ, angles of incidence α and diffraction β, and radii of curvature R (meridional) and ρ (sagittal).

2.2. Meaning of the Coefficients

Term	Name	Effect when non-zero
F_{00}	Central path length	Constant (irrelevant for focusing)
F_{01}	Grating equation term	Selects wavelength; always present
F_{20}	Defocus (meridional)	Shifts focus along optical axis
F_{02}	Defocus (sagittal)	Sagittal focus mismatch
F_{30}	Primary coma	Asymmetric blur at exit slit
F_{21}	Mixed (coma–sagittal)	Couples sagittal and meridional aberrations

In practice, the key design goals are:

- $F_{20} = 0$: meridional focus at the exit slit position.
- $F_{30} = 0$: coma-free operation (often achievable at one energy only).

2.3. Grating Equation and Resolving Power

The grating equation in diffraction order m is:

$$m\lambda = d\,(\sin\alpha - \sin\beta), \tag{2}$$

where d is the groove spacing, α is the angle of incidence (positive above normal), and β is the diffraction angle (negative in the SGM convention used here, where $m = -1$ in SHADOW notation corresponds to the physically relevant first order).

The theoretical resolving power is determined by the exit-slit-limited resolution:

$$\boxed{\mathrm{RP} = \frac{\lambda}{\Delta\lambda} = \frac{r'\,m\cos\beta}{d\cdot\Delta_{\mathrm{slit}}},} \tag{3}$$

where r' is the grating-to-slit distance and Δ_{slit} is the FWHM of the beam at the slit plane.

2.4. Image Size: Sanity Check Formula

A key consistency check in all simulations is the predicted image size at the exit slit, obtained by geometrical demagnification:

$$\Delta_z = 2.35\,\sigma_z \frac{\cos\alpha}{\cos\beta}\frac{r'}{r} \quad \text{(vertical/meridional plane)}, \tag{4}$$

$$\Delta_x = 2.35\,\sigma_x \frac{r'}{r} \quad \text{(horizontal/sagittal plane)}, \tag{5}$$

where σ_z, σ_x are the source RMS sizes and r, r' are the source-to-grating and grating-to-image distances. Agreement between the ray-tracing result and Eqs. (4)–(5) confirms that the optical system is properly focused.

Fig. 3

Diagram illustrating the demagnification geometry: source of size σ_z at distance r, grating with angles α and β, and image of size Δ_z at distance r', showing the $(\cos\alpha/\cos\beta)$ asymmetry factor.

3. Spherical Grating Monochromator (SGM)

3.1. Optical Layout

The SGM is the simplest SXR monochromator: a single **concave spherical grating** works simultaneously as the dispersing and focusing element. The geometry is defined by:

- Source-to-grating distance r.
- Grating-to-slit distance r' (movable mechanically).
- Bragg-like incidence angle α and diffraction angle β, both measured from the grating normal.
- Diffraction order $m = -1$.

The defocus condition $F_{20} = 0$ gives the **grating equation for focusing**:

$$\frac{\cos^2\alpha}{r} + \frac{\cos^2\beta}{r'} = \frac{\cos\alpha + \cos\beta}{R}, \tag{6}$$

where R is the grating radius of curvature. At each energy, α and β change (coupled through Eq. (2)), so the focal distance r' also changes. Therefore, **the SGM requires a movable exit slit** to track the focus.

Fig. 4

SGM optical layout: source at distance r, single spherical grating (radius R), and movable exit slit at distance $r'(E)$. Indicate the angles α and β, and the energy-dependent slit travel range.

SGM design rule: Because $r'(E)$ is energy-dependent, a fixed exit slit position can only be optimal at one energy. At all other energies, coma and defocus degrade the resolution. The exit slit must be translated to follow the focus.

3.2. Ray-Tracing Example at 1000 eV

File: `01_SGM_1000eV.ows`

Parameters: $r = 30\,\mathrm{m}$, $r' \approx 9.9\,\mathrm{m}$, $\alpha = 87.3$ř, $\beta = 89.1$ř, diffraction order $m = -1$.

Results:

- Predicted vertical image size (sanity check): $\Delta_z = 2.35\,\sigma_z\,(\cos\alpha/\cos\beta)\,(r'/r) \approx 36\,\mu\mathrm{m}$.
- Predicted horizontal image size: $\Delta_x \approx 217\,\mu\mathrm{m}$.
- Achieved resolving power: RP $\approx 18\,000$ at 1000 eV with 55 meV FWHM energy resolution.
- Ray-tracing confirms both sanity check values, validating the optical layout.

Slit scan: Resolution is *source limited* for slit openings $\lesssim 20\,\mu\mathrm{m}$; for larger openings ($\gtrsim 50\,\mu\mathrm{m}$) the resolution is *slit limited*, degrading linearly with slit width.

Fig. 5

SHADOW ray-tracing results for SGM at 1000 eV. Left: footprint and spot diagram at the exit slit plane ($\Delta_z \approx 36\,\mu m$, $\Delta_x \approx 217\,\mu m$). Right: resolving power vs. slit opening, showing source-limited and slit-limited regimes with the transition near $20\,\mu m$.

3.3. Ray-Tracing Example at 1600 eV

File: `02_SGM_1600eV.ows`

Parameters: $\alpha = 87.63$ř, $\beta = -88.77$ř. Source: $S_x = 280\,\mu$m, $S_z = 154\,\mu$m, $S'_x = 18\,\mu$rad, $S'_z = 14\,\mu$rad.

Key finding — focus shift with energy:
The sanity-check formula predicts $\Delta_z \approx 20\,\mu$m at the nominal slit position, but the ray-tracing footprint shows the beam is *not at focus* at 1600 eV if the exit slit remains at the 1000 eV position. Moving the exit slit to $r' = 11\,046$ mm (satisfying $F_{20}(1600\,\text{eV}) = 0$) restores focus and gives $\Delta_z \approx 22\,\mu$m, consistent with the sanity check.

Conclusion: This example confirms that the SGM requires slit translation with energy. The exit slit movement *corrects defocus but does not correct coma.*

Fig. 6

SGM at 1600 eV: comparison of slit-plane footprints (a) at the fixed 1000 eV slit position (defocused, asymmetric) and (b) after slit translation to $r' = 11\,046$ mm (in focus, $\Delta_z \approx 22\,\mu$m). The residual coma asymmetry remains in both cases.

4. Plane Grating Monochromator (PGM)

4.1. Key Properties of a Plane Grating

A plane grating (without variable line spacing) **does not focus**. Its role is purely dispersive. The focusing is delegated entirely to curved mirrors upstream and downstream of the grating. The important consequence is a **fixed focal plane**: the image distance is independent of the grating rotation.

The grating magnification factor is defined as:

$$c = \frac{\cos\beta}{\cos\alpha}, \tag{7}$$

and the grating equation becomes:

$$1 - \frac{m\lambda}{d\sin\beta} = \frac{1}{c^2}. \tag{8}$$

The parameter c controls the angular compression/expansion between the collimated beam and the diffracted beam, and thus the **grating magnification** c^{-1} can be tuned by rotating the plane mirror that delivers the collimated beam.

Fig. 7

Schematic of the plane grating geometry. Show the plane mirror (angle θ_M) delivering the collimated beam at incidence angle α to the plane grating, and the diffracted beam at angle β being focused by the downstream cylindrical mirror. Label the magnification factor $c = \cos\beta/\cos\alpha$.

4.2. Collimated PGM: Vertical Deflection Layout

File: `03_CollimatedPGM1000eVc2_Vertical.ows`

Layout (distances from source): Toroidal mirror at 9 m → plane mirror + plane grating at 28 m → cylindrical mirror at 30 m → exit slit at 31 m.

Optical roles:

- **Toroidal mirror:** collimates the beam in the meridional (vertical) plane; focuses sagittally (horizontally) at the exit slit.
- **Plane mirror + plane grating:** deflect in the vertical plane; the plane mirror rotation controls c and hence the wavelength.
- **Cylindrical mirror:** focuses the collimated, dispersed beam onto the *fixed* exit slit.

Parameters at 1000 eV, $c = 2$: $\alpha = 88.5$ř, $\beta = 87.1$ř; plane mirror at 87.8ř incidence.

Aberrations:

- **Coma** is present in the vertical plane because the collimated beam does not enter the grating exactly at normal incidence (residual F_{30}).
- Image size at slit: $\sim 5.8\,\mu$m without slope errors.
- **Slope errors** on the toroidal mirror broaden the slit image significantly. The simulation shows that slope errors on the upstream toroid are the dominant resolution-limiting factor in this configuration.

Fig. 8

Vertical collimated PGM layout ($c = 2$, 1000 eV). Top: beamline schematic with distances (toroid at 9 m, plane mirror + grating at 28 m, cylinder at 30 m, slit at 31 m). Bottom: ray-tracing spot at the exit slit showing coma aberration ($\sim 5.8\,\mu$m ideal) and the broadened profile with slope errors on the toroid.

Fig. 9

Effect of slope errors on the toroidal mirror in the vertical PGM. Comparison of slit images: (a) without slope errors (ideal, $\sim 5.8\,\mu$m) and (b) with realistic r.m.s. slope errors, demonstrating the dominant contribution of the toroid to the resolution budget.

4.3. Follath Collimated PGM: Horizontal Deflection

The **Follath PGM** (R. Follath and F. Senf, NIM A 390, 388, 1997) uses **horizontal** deflection by the plane mirror and grating. The key advantage is that, in the horizontal plane, slope errors on the toroidal mirror are much less critical because the sagittal focusing is less sensitive to meridional slope errors.

File: `04_CollimatedPGM1000eVc2.ows`

Same layout as above, but mirror–grating system rotates in the horizontal plane.

Key results at $c = 2$, 1000 eV:

- **No coma:** the horizontal geometry naturally satisfies $F_{30} \approx 0$ due to the relative rotation between the plane mirror and the grating in the xz-plane.
- Image size: $\sim 5.8\,\mu$m (same as vertical case without aberrations, but now robust against mirror slope errors).
- The **"forgiveness factor"** $\times\theta$: slope errors on the toroidal mirror have a much smaller effect in the horizontal geometry.
- Simulation with and without slope errors confirms that the Follath layout is significantly more tolerant to mirror figure errors.

Effect of increasing c: At $c = 5$ (file `05_CollimatedPGM1000eVc5.ows`), $\alpha = 89.5$ř, $\beta = 87.4$ř. The **astigmatic coma** grows, scaling as $\sim 5.8/(c/2) = 2.32\,\mu$m for ideal optics. With slope errors, the image quality degrades further.

Fig. 10

Follath horizontal PGM ($c = 2$, 1000 eV). Left: beamline plan view showing horizontal deflection by the plane mirror and grating. Centre and right: SHADOW footprints at the exit slit without and with toroid slope errors, demonstrating the "forgiveness factor" — the slope errors have negligible effect in this geometry.

Fig. 11

Astigmatic coma growth with increasing magnification factor c in the Follath PGM. Slit images at $c = 2$ and $c = 5$ (file `05_CollimatedPGM1000eVc5.ows`), showing the $5.8/(c/2)$ scaling of the coma contribution.

5. Focusing Variable Line Spacing PGM (FVLS-PGM)

5.1. Principle

The **FVLS-PGM** (R. Reininger and A. R. B. de Castro, NIM A 538, 760, 2005) addresses the two main limitations of the collimated PGM:

1. **Heat-load-induced deformations** of the plane mirror degrade the resolution by introducing a virtual source shift.
2. **Coma** cannot be simultaneously cancelled at all energies with a fixed line-spacing grating.

The solution is to use a **variable line spacing** (VLS) grating, where the groove density varies along the grating surface as:

$$\sigma(w) = \sigma_0 + b_2\, w + b_3\, w^2 + \cdots\,, \tag{9}$$

with $\sigma_0 = 1/d_0$ the central line density. The coefficients b_2 and b_3 are chosen to:

- b_2: solve the defocus equation $F_{20} = 0$ at all wavelengths simultaneously, by illuminating the grating at the correct angle of incidence (note: in the FVLS design, c is *not* a free parameter — it is determined by b_2).
- b_3: cancel coma $F_{30} = 0$ at one selected wavelength; coma is then acceptably small over the full energy range.

FVLS design parameters (example at central energy 600 eV, $c = 2$):

$$b_2 = 1.44 \times 10^{-4}\,\mathrm{mm}^{-1},$$
$$b_3 = 1.3 \times 10^{-8}\,\mathrm{mm}^{-2}.$$

Coma is zeroed at 600 eV; solving coma at one wavelength is sufficient because the residual coma at other energies remains small.

Fig. 12

VLS grating schematic showing the variable groove spacing $d(w) = 1/\sigma(w)$ as a function of position w along the grating surface. Indicate the central density σ_0 and the quadratic variation controlled by b_2 and b_3.

5.2. Optical Layout

File: `06_FVLSPGM.ows`

Distances: Source → heat absorber → plane mirror (28 m) → VLS grating (28.3 m) → slit plane (38.3 m) → refocusing mirror → sample (48.3 m).
Parameters at 1000 eV: $\alpha = 88.5$ř, $\beta = 87.1$ř (same $c = 2$ condition).

Key feature: There is no upstream collimating mirror. The VLS grating itself provides the meridional focusing at the exit slit, and the horizontal focus is provided after the slit by a separate refocusing mirror.

Fig. 13

FVLS-PGM beamline layout. Show the heat absorber, plane mirror (28 m), VLS grating (28.3 m), exit slit (38.3 m), and post-slit refocusing mirror. Annotate the distances and the absence of an upstream collimating mirror compared to the Follath PGM.

5.3. Heat Load and Correction

The crucial advantage of the FVLS-PGM emerges when heat-load deformations of the plane mirror are considered:

Heat load simulation: The plane mirror absorbs ~ 35 W of total power. The resulting thermo-mechanical deformation produces a maximum surface height deviation of $\sim 0.40\,\mu$m and slope errors up to $\sim 15\,\mu$rad peak. The effective radius of curvature induced is $R \approx -30$ km (very weakly concave). This shifts the virtual source position to ~ 28.52 m.

Without correction: the deformed mirror creates a defocused, asymmetric image at the slit — resolving power degrades substantially.

Correction strategy: Because the FVLS grating corrects defocus at all energies by adjusting the angle of incidence, a *small angular correction* to α and β is sufficient to

restore the focus:

$$\alpha: \ 88.52843\check{r} \to 88.53789\check{r}, \quad \beta: \ 87.05389\check{r} \to 87.05860\check{r}.$$

This is equivalent to accounting for the shifted virtual source position in the VLS design equations.

Result: After angular correction, the ray-tracing shows full recovery of the nominal slit image — the FVLS-PGM **self-corrects for heat-load deformations** with only a minor adjustment of the mirror–grating angles.

Reference: R. Reininger, NIM A 649, 139 (2010).

Fig. 14

Heat load deformation of the plane mirror (35 W absorbed power). Left: surface height deviation (peak $\sim$ 0.40 μm) and slope error profile (peak $\sim$ 15 μrad) vs. mirror length. Right: induced virtual-source shift to $\sim$ 28.52 m from the nominal 28 m position.

Fig. 15

FVLS-PGM ray-tracing at the exit slit under three conditions: (a) no heat load (reference); (b) with heat load, uncorrected (defocused, RP degraded); (c) with heat load, corrected by angular adjustment ($\Delta\alpha = +0.00946\check{r}$, $\Delta\beta = +0.00471\check{r}$) — full RP recovery.

Limitation of the FVLS-PGM: Because c is fixed by the VLS coefficients (b_2), the grating magnification *cannot* be changed during operation. The Follath collimated PGM retains the freedom to vary c (and hence the resolving power vs. flux trade-off), but at the cost of reduced heat-load tolerance.

6. Comparison of Monochromator Types

Property	**SGM**	**Vertical PGM**	**Follath PGM**	**FVLS-PGM**
Exit slit position	Movable (tracks focus)	Fixed	Fixed	Fixed
Coma	Present	Present	Eliminated (hor. geom.)	Zeroed at one λ
Slope error sensitivity	Moderate	High (toroid)	Low (forgiveness factor)	Low
Heat load correction	Slit movement	Slit movement	Slit movement	Angle correction
Variable magnification c	N/A	Yes	Yes	No (fixed by VLS)
Focusing element	Grating only	Toroid + cylinder	Toroid + cylinder	VLS grating

Fig. 16

Side-by-side comparison of the four monochromator layouts drawn to the same scale, highlighting the presence/absence of the upstream collimating mirror, direction of deflection (vertical vs. horizontal), and the position of the exit slit relative to the grating.

7. Summary and Conclusions

The lecture by Reininger establishes a unified theoretical framework — the optical path length expansion of Eq. (1) and Fermat's principle — that underpins all grating monochromator designs for soft X-rays. The main conclusions are:

1. **The SGM** is conceptually simple but operationally demanding: the exit slit must track the focus with energy, and coma cannot be corrected independently.
2. **The collimated PGM** decouples focusing from dispersion. The *vertical* layout is coma-affected and sensitive to toroidal mirror slope errors. The *Follath horizontal* layout eliminates coma and is far more tolerant to slope errors (the "forgiveness factor").
3. **The FVLS-PGM** is the most robust design: VLS coefficients simultaneously cancel defocus at all energies and coma at the design energy, and the grating angles can be

slightly adjusted to compensate heat-load deformations of the plane mirror without moving the exit slit.

4. **OASYS/SHADOW** provides the numerical environment to implement all these designs and verify them against the analytical sanity checks (image size, resolving power). The formulas for all monochromator types discussed here are included in the OASYS widget library.

Fig. 17

Summary figure: resolving power RP *vs. photon energy for the four monochromator types simulated in this work, plotted on a common axis to allow direct performance comparison. Each curve corresponds to the optimal operating condition of the respective instrument.*

Glossary of Symbols

Symbol	Definition
F	Optical path length function
F_{ij}	Coefficients of the OPL expansion (i: meridional order, j: sagittal order)
w	Coordinate along the groove direction (sagittal)
ℓ	Coordinate across the grooves (meridional)
α	Angle of incidence on the grating (from grating normal)
β	Angle of diffraction
m	Diffraction order
d	Groove spacing (at grating centre)
λ	Wavelength
r	Source-to-grating distance
r'	Grating-to-image (exit slit) distance
R	Grating radius of curvature (SGM)
c	Magnification factor ($= \cos\beta/\cos\alpha$)
RP	Resolving power ($= \lambda/\Delta\lambda$)
S_x, S_z	Source size (horizontal, vertical) RMS
S'_x, S'_z	Source divergence (horizontal, vertical) RMS
b_2, b_3	VLS grating coefficients (line density gradient, curvature)
σ_0	Central groove density of VLS grating

References

1. R. Follath and F. Senf, *New plane grating monochromators for third generation synchrotron radiation light sources*, Nucl. Instrum. Methods Phys. Res. A **390**, 388 (1997). doi:10.1016/S0168-9002(97)00401-4

2. R. Reininger and A. R. B. de Castro, *The Focusing VLS-PGM: A new monochromator for soft X-rays*, Nucl. Instrum. Methods Phys. Res. A **538**, 760 (2005). doi:10.1016/j.nima.2004.09.007

3. R. Reininger, *Correction of heat-load-induced deformations in soft X-ray plane grating monochromators*, Nucl. Instrum. Methods Phys. Res. A **649**, 139 (2010).

How to Replace Placeholders with Real Figures

Each placeholder in this document is produced by the command:

```
\placeholder[width]{height}{Fig. N}{Description of intended content}
```

To replace a placeholder with a real figure, comment out the `\placeholder` line and insert a standard LaTeX figure block in its place:

```
% \placeholder{5cm}{Fig.~5}{...description...}
\begin{figure}[H]
  \centering
  \includegraphics[width=0.85\linewidth]{fig05_SGM_1000eV_raytracing.pdf}
  \caption{SHADOW ray-tracing results for SGM at 1000\,eV. ...}
  \label{fig:SGM_1000eV}
\end{figure}
```

Recommended image formats (in order of preference): `.pdf`, `.eps`, `.png`. Place all figure files in the same directory as the `.tex` source, or adjust the path in `\includegraphics` accordingly.